

Selfless Traffic Routing with Multi-Agent PPO in SUMO: When Does Sacrificing Help?

Anonymous submission

Abstract

Most navigation systems route each vehicle along its own shortest path; when many vehicles make the same locally optimal choice, the roads they converge on become bottlenecks and total travel time rises, opening the gap between the selfish (Wardrop) equilibrium and the system optimum known as the price of anarchy. *Selfless routing* asks some vehicles to accept a longer personal route when doing so lowers travel time for the whole fleet. We study whether cooperative multi-agent reinforcement learning can learn this behavior in a realistic simulator: Multi-Agent Proximal Policy Optimization (MAPPO) with centralized training and decentralized execution, a shared actor that scores up to four complete candidate routes, and a difference-reward team term measured in seconds of travel time, with a congestion-gated incentive that pays for a detour only when it relieves a real jam. Trained on a congested New York City corridor (450 controlled and 150 background vehicles) and evaluated against a Dijkstra shortest-path baseline on 20 disjoint held-out scenarios, the deployed greedy policy completes 100% of vehicles and reduces mean fleet travel time from 596.6 s to 534.3 s (−10.4%, winning 15/20 scenarios, paired Wilcoxon $p=0.02$), with lower 90th-percentile travel time and fewer deadline misses; it executes genuine load-spreading detours (38% non-shortest choices), and the improvement grows with congestion ($r = -0.57$), reaching −27% on a more severely congested block. Ablating the congestion-aware detour guard, which reverts only pile-on detours, gains a further ∼3% concentrated entirely at high congestion, and a fixed-demand penetration sweep shows the typical gain growing monotonically with the controlled share of the fleet. The value of selfless routing tracks the network’s price of anarchy, so it is largest exactly where congestion is worst.

1 Introduction

Traffic congestion is a system-level phenomenon, yet most routing systems attack it as a collection of independent shortest-path problems: each vehicle is guided along the route that is fastest for itself, taking the current network as fixed. When a large number of vehicles make that same locally optimal choice, the roads they share fill up, and the

locally optimal path becomes a bottleneck that raises travel time for everyone. Classical traffic theory formalizes this as the gap between the *selfish* routing outcome (the Wardrop user equilibrium, in which no driver can improve their own time by switching routes (Wardrop 1952)) and the *system optimum* that minimizes total travel time. The ratio between them is the *price of anarchy*, and it can be strictly greater than one (Roughgarden and Tardos 2002); the Braess paradox is the extreme case, where adding capacity under selfish routing makes everyone slower (Braess, Nagurney, and Wakolbinger 2005).

The *selfless traffic routing* (STR) model of Dai et al. (2019) proposes the constructive alternative: allow some controlled vehicles to accept a slightly longer route when that choice reduces the average travel time of the fleet, subject to individual constraints. This reframes routing as a cooperative decision problem rather than a per-vehicle optimization. The open question is whether a learned policy can discover *when* such sacrifices pay off, and by how much, in a realistic traffic environment rather than an idealized graph.

This paper studies that question with a cooperative multi-agent reinforcement-learning (MARL) controller in the Eclipse SUMO microscopic simulator (Lopez et al. 2018). We model each controlled vehicle as an agent that periodically chooses among a small set of complete candidate routes to its destination, and we train the shared policy with Multi-Agent Proximal Policy Optimization (MAPPO) (Yu et al. 2022), a centralized-training / decentralized-execution (CTDE) actor–critic method. Crucially, cooperation is not left to emerge on its own: each agent’s reward includes a share of the fleet’s congestion cost, expressed *in seconds of travel time*, so that the policy can trade its own delay against fleet delay on a common scale. We compare the learned controller against a Dijkstra shortest-path baseline (Dijkstra 1959) on the same generated demand.

Although this study is carried out in simulation, the ultimate application is on actual vehicles, where the controller’s inputs (local observations, shared congestion summaries, and committed route intents) must travel over a vehicular network. Emerging vehicle-to-everything (V2X) communication makes this practical: IEEE 802.11bd, the next-generation V2X amendment (IEEE 2022), provides the connectivity and bandwidth required by intelligent transportation systems of this kind (Iliopoulos et al. 2025).

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1.1 Research Question

Can a cooperative MAPPO routing policy reduce fleet travel time relative to shortest-path routing, and under what traffic conditions do selfless (non-shortest) route choices actually help?

The second half of the question is as important as the first: our results show that the answer depends sharply on how congested the scenario is.

1.2 Contributions

- We formulate selfless routing as a cooperative Dec-POMDP over route choices and solve it with a CTDE MAPPO policy whose actor scores complete candidate routes and whose reward (including a *difference-reward* team term (Wolpert and Tumer 2001)) is expressed entirely in travel-time units, avoiding an arbitrary congestion-vs-time exchange rate.
- We evaluate the trained policy against Dijkstra on 20 disjoint held-out congested scenarios, reporting an honest, reproducible outcome: the deployed greedy policy reduces mean fleet travel time by 10.4% (15/20 wins, paired Wilcoxon $p=0.02$) at 100% completion, executing genuine load-spreading detours (38% non-shortest choices) rather than only replanning, with gains that grow with congestion. A fixed-demand penetration sweep shows the typical per-scenario gain growing monotonically as the controlled share of the fleet rises from 10% to 100%.
- Through a guardrail ablation we show that the extra benefit of executing every learned detour (rather than reverting the pile-on ones) is concentrated almost entirely in heavy congestion and vanishes in light traffic, a direct empirical link between the benefit of selflessness and the network’s price of anarchy.
- We document the failure modes that remain (a rare gridlock scenario, a slow upward drift in the fleet detour rate, and a weak per-update learning signal on a dense grid) and the role of a recalibrated congestion-aware guardrail that reverts only detours which relieve nothing.

2 Related Work

2.1 Selfish routing and the price of anarchy

The study of routing as a game predates learning-based methods. Wardrop’s principles distinguish the user equilibrium, where every used route between an origin and destination has equal and minimal cost to the individual, from the system optimum that minimizes total cost (Wardrop 1952). Roughgarden and Tardos (2002) quantified how far apart these can be, proving bounds on the price of anarchy for selfish routing, and the Braess paradox shows the equilibrium can be counter-intuitively bad (Braess, Nagurney, and Wakolbinger 2005). The gap is not merely theoretical: Zhang et al. (2018) estimated the price of anarchy from real traffic data on the Eastern-Massachusetts road network and proposed data-driven strategies for reducing it, confirming that deployed networks operate measurably above the system optimum. These results motivate selfless routing directly: the potential benefit of asking vehicles to sacrifice is exactly the

price-of-anarchy gap, which is large only when the network forces contention onto scarce links. Our experiments make this dependence concrete: selfless detours help precisely in the saturated scenarios where the gap is wide, and not otherwise.

2.2 The selfless traffic routing model

Dai et al. (2019) introduced the selfless driving model and argued that vehicular routing should be evaluated by network-level average travel time rather than individual shortest paths. Their formulation permits a controlled vehicle to take a locally suboptimal route when doing so benefits the group. We adopt this objective but replace their heuristic controller with a learned cooperative policy, and we ground the trade-off in a common travel-time unit so the sacrifice decision is made on principled terms.

2.3 Reinforcement learning and MAPPO

Reinforcement learning is a natural fit for sequential routing, where actions change the state that later actions face (Mnih et al. 2015). Proximal Policy Optimization (PPO) provides a stable on-policy actor-critic update via a clipped surrogate objective (Schulman et al. 2017), and Generalized Advantage Estimation (GAE) reduces the variance of the advantage signal it optimizes (Schulman et al. 2016). In the multi-agent setting, the environment becomes non-stationary from any single agent’s view. The centralized-training / decentralized-execution paradigm addresses this by giving the critic access to global information during training while the actor remains local (Lowe et al. 2017). MAPPO applies PPO under CTDE with a centralized value function and has proven surprisingly strong on cooperative benchmarks (Yu et al. 2022); we use it here because our reward is cooperative and our deployment must be decentralized (each vehicle routes from its own view).

2.4 Credit assignment in cooperative MARL

A cooperative reward shared identically by all agents makes it hard to tell which agent’s action helped, because each agent’s contribution is buried in a large common signal. Difference rewards address this by crediting each agent with its marginal effect on the global objective (Wolpert and Tumer 2001), and counterfactual multi-agent policy gradients (COMA) estimate that marginal effect with a centralized critic (Foerster et al. 2018). Our team-reward term is a leave-one-out difference reward in travel-time units: each agent pays for its own slowness relative to the live fleet average, so the external, demand-driven congestion level, which no single vehicle controls, cancels out and the learnable signal is sharpened.

2.5 Simulation environment

We use Eclipse SUMO as the microscopic (vehicle-level) traffic simulator (Lopez et al. 2018; Krajzewicz et al. 2012). SUMO models car-following, lane changing, junctions, and traffic lights, which makes the evaluation considerably more realistic than routing on a static weighted graph, but also makes control harder: a route that looks good at the graph level is only useful if the vehicle can physically execute it

under lane-feasibility, route-continuity, and commit-window constraints. Dijkstra’s algorithm (Dijkstra 1959) remains the appropriate baseline, not as a weak straw man but as a strong, deterministic controller that respects the road graph and never over-commits, which is why it is competitive in exactly the low-congestion regime where selfless routing has nothing to gain.

3 Method

3.1 Problem Formulation

We control a fleet of N vehicles on a fixed SUMO road network while a background population of uncontrolled vehicles shares the roads. Each controlled vehicle i is an agent that, at a set of route-decision points along its trip, chooses a complete route to its destination. SUMO handles all low-level dynamics (car-following, lane changing, junctions, and signals), so the policy only sets high-level route intent. The fleet objective is to minimize average travel time,

$$J(\pi) = \frac{1}{N} \sum_{i=1}^N T_i, \quad (1)$$

where T_i is the realized travel time of vehicle i under the shared policy π . This is a cooperative, partially observed problem: each agent sees only a local view of the network but its route choice affects the congestion that other agents experience. We treat it as a decentralized POMDP and solve it with CTDE MAPPO, so that a fleet-wide signal can guide learning while deployment stays per-vehicle.

The tension the policy must resolve is precisely the price-of-anarchy gap of Section 2: the route that minimizes T_i for one vehicle need not minimize J for the fleet. A useful policy learns when to keep a vehicle on its shortest feasible path (most of the time) and when to route it away from a forming bottleneck for the group’s benefit.

3.2 Route-Candidate Action Space

Rather than asking the policy to pick a turn (“left/right”) at each junction, a route-candidate generator proposes up to $K=4$ *complete* routes from the vehicle’s current edge to its destination. Index 0 is always the congestion-aware shortest route; indices 1–3 are diverse detours produced by density-penalized Dijkstra searches. The actor’s job is to choose one of these routes, so a “selfless detour” simply means choosing an index other than 0. Each candidate carries an 11-dimensional feature “scorecard” describing, among other things, its average and worst-case congestion, its first-edge congestion, the extra seconds it costs relative to the shortest route (the ETA sacrifice), and how much less congested it is than the shortest route (the relief). Because the reward, the actor, and the guardrails all read the same scorecard, a trade-off assessed by one component is understood identically by the others.

3.3 CTDE Actor and Critic

The **actor** is a shared per-candidate scorer: a small multilayer perceptron (two hidden layers of 256 and 128 units, ReLU) takes the vehicle’s local observation concatenated with one

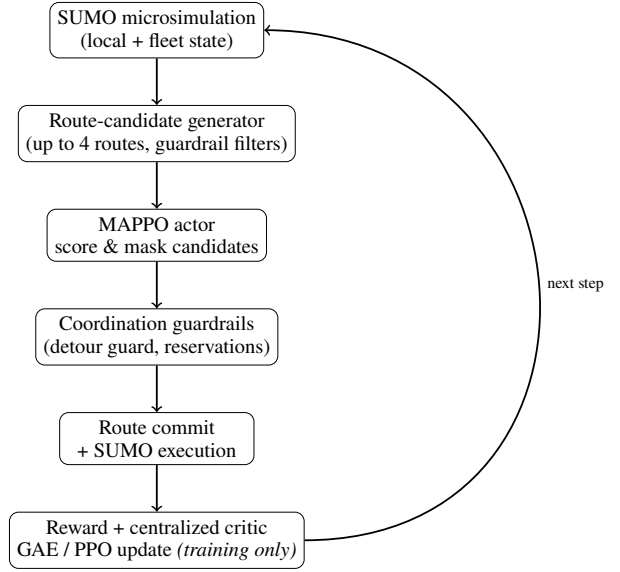


Figure 1: The STR-SUMO learning and deployment pipeline. Every stage speaks a common travel-time / density vocabulary. The dashed portion (reward and critic) runs only during training; at deployment the actor and guardrails alone drive each vehicle.

candidate’s features and emits a scalar score; the same sub-network scores all K candidates, and invalid candidates are masked to -10^9 before a softmax over scores. This weight-sharing lets the actor generalize across a variable, unordered candidate set and keeps deployment per-vehicle. The local observation encodes the current and destination edges, feasibility masks, lane and commit-window state, route-difficulty estimates, local density summaries, and the per-candidate features.

The **critic** is used only during training. It concatenates the local observation with an 18-dimensional *central* observation summarizing the whole fleet’s state (aggregate densities, completion, congestion pressure) and predicts a scalar value. This is the CTDE split (Lowe et al. 2017): the critic “peeks” at global information to give lower-variance value estimates, while the actor never does, so the trained policy runs decentralized. The critic changes learning stability, not the objective; the cooperative incentive must come from the reward.

3.4 Learning Algorithm

The policy is trained with MAPPO (Yu et al. 2022): a clipped PPO surrogate for the actor (Schulman et al. 2017) and a squared-error value loss for the critic, with an entropy bonus for exploration and separate Adam optimizers (Kingma and Ba 2015) for the two networks. Advantages are computed per vehicle trajectory with GAE (Schulman et al. 2016) ($\gamma=0.995$, $\lambda=0.95$); the high discount is needed because trips span roughly two thousand simulation steps and a route choice must be credited for congestion it causes far downstream. Two practical components proved important. First, because episode returns swing by more than an order of

Table 1: MAPPO Training Configuration

Group	Parameter	Value
Actor/critic	hidden layers	[256, 128], ReLU
	candidate routes K	4
	central obs dim	18
Optimization	actor / critic LR	$3 \times 10^{-4} / 1 \times 10^{-3}$
	γ / GAE λ	0.995 / 0.95
	PPO clip ϵ	0.05
	epochs / minibatch	4 / 512
	target KL	0.015
	value-target norm.	on
Exploration	entropy coef. (start→end)	0.15 → 0.10
Objective	team-reward α	1.0
	team-reward mode	difference
	detour refund fraction	0.6

magnitude with the demand seed, the critic regresses on *normalized* value targets (running mean/variance) while advantages stay in real reward units, keeping the value loss well-conditioned. Second, entropy is annealed from 0.15 to 0.10 so the policy explores detours early and settles later, with PPO updates stopped early when the approximate KL exceeds 0.015. Key hyperparameters are in Table 1.

3.5 Reward: Cooperation in Travel-Time Units

Every term of the reward is measured in seconds so trade-offs need no arbitrary conversion factor. Each simulation step charges a small base time cost (0.07 per step), plus deliberately smaller costs for occupying above-average-density roads and for the congestion a vehicle imposes on others; reaching the destination pays +50 and a teleport (SUMO’s last resort for a stuck vehicle) costs −40. The cooperative term is the key addition. Let ρ_i be vehicle i ’s normalized slowness (0 at free flow, 1 at standstill) and $\bar{\rho}$ the live-fleet mean. With weight $\alpha \in [0, 1]$ and scale κ , agent i pays over an elapsed interval Δt

$$c_i^{\text{team}} = \alpha \kappa (\rho_i - \bar{\rho}) \Delta t. \quad (2)$$

This is a leave-one-out difference reward (Wolpert and Tumer 2001): because the fleet-mean of $(\rho_i - \bar{\rho})$ is zero, the external, demand-driven congestion level cancels, and each agent is credited only for how its own route choice fares relative to the fleet: a vehicle that routed onto a free-flowing alternative is rewarded, one that piled into a jam pays. We use $\alpha=1$ (fleet delay counts as much as personal delay) with the difference mode throughout; $\alpha=0$ recovers a purely selfish objective.

Congestion-gated detour credit. A worthwhile detour still costs the extra seconds it adds, so without a counterweight a mean objective drives the policy back to the shortest path. We therefore add a small detour-shaping term, read from the same scorecard and *gated by congestion*: a gate $g = \max(\bar{d} - \delta, 0)$ on the baseline density $\bar{d}$ is near zero in light traffic (so the policy is never paid to detour for nothing) and grows with congestion, where it lowers the relief bar a detour must clear and boosts *genuine* measured relief (a lateral move onto an equally dense road earns nothing). A partial (60%) time refund leaves a residual cost, so only

clearly relieving detours pay: the per-decision value is positive under congestion and negative in light traffic, as the deployment results confirm.

3.6 Guardrails and the Saturation Veto

Hand-written guardrails keep the reward focused on travel time by removing obviously bad options rather than teaching basic safety through penalties. Pre-decision filters prune candidate routes that would loop, re-enter a dead end, or drive straight into a known jam; a “pending” state machine distinguishes healthy queueing (e.g. at a red light) from a genuine stall so the reward is not charged for normal waiting. Two coordination guardrails address a specific failure in heavy congestion, where independently chosen detours can pile onto the same alternative and make it worse:

- The **detour guard** (deployment only): after the actor picks a detour, if the alternative is itself near capacity *and* the relief it claims is within measurement noise, the choice is reverted to the shortest route—so the guard blocks only *pointless pile-on* detours that add distance without relieving anything, while a near-capacity detour showing *genuine* relief is allowed through. It judges crowdedness in *absolute* terms, not the relief the policy reacted to, because when everything is busy a road can look like relief yet have no room left. (A blanket veto whenever the whole network is saturated, which we tried first, instead reverts essentially every detour and hides the learned policy, so we do not use it.) It runs only at deployment, since overriding an executed route during training would break the on-policy assumption.
- The **route reservation** (training and deployment): a committed route books its leading edges in a decaying field, so the candidate generator scores later deciders against effective (live + reserved) density and a simultaneous rush becomes an orderly one-at-a-time spread. It affects only what candidates look like, never the reward or the true measurements.

The detour guard is the subject of our ablation in Section 5.3: because it now reverts only the pile-on subset of detours, it costs a few percent of mean fleet gain in exchange for a small robustness margin on the worst gridlock scenario.

4 Experimental Setup

4.1 Network and Demand

Experiments use a corner of the New York City street grid, cut from OpenStreetMap (OpenStreetMap contributors 2026) and converted to a SUMO network with SUMO’s OSM import tooling (Lopez et al. 2018): 132 drivable edges, 360 lanes, 90 junctions (53 signalized), and roughly 27,000 lane-meters. To create genuine congestion (a precondition for selfless routing to have anything to gain), we use a shared-destination demand pattern (ranged origins routed to one common destination edge) with 450 controlled and 150 background vehicles released at a 0.5 s interval. This concentrates flow into a corridor and produces sustained congestion (frequent teleports, fleet travel times from a few hundred to over 1500 s across seeds) without permanent gridlock. A

dispersed origin–destination pattern leaves the grid at a few percent of capacity, where routing cannot matter.

4.2 Protocol, Baselines, and Metrics

The policy is trained for 150 episodes. Every 25 episodes it is evaluated frozen (no learning) on 30 seeds (6000–6029) against a Dijkstra baseline generated from the identical demand; the best checkpoint is selected on the *greedy* deployment tail (p90 first, then mean), which is the configuration actually shipped. Because a single greedy argmax pass can hide a still-improving policy, the frozen eval also runs a seeded, averaged stochastic pass as a learning-detectability diagnostic (it does not drive selection). We then *deploy* the selected checkpoint on a *separate, disjoint* block of 20 held-out scenarios (seeds 4010–4029)—never seen during training or selection—and compare it head-to-head with Dijkstra on the same generated vehicles. We report mean fleet travel time, median (p50) and p90 travel time, tail-completion gap, the p95/p50 ratio, deadline misses, completion rate, and per-scenario win rate. Unless noted, the MAPPO policy is deployed *greedily* (argmax over candidate scores), which is deterministic and, as Section 5.5 shows, outperforms stochastic sampling in this regime. Each controller is run in a fresh simulation with independent network state, so no traffic leaks between runs.

Finally, we sweep the *controlled-vehicle penetration*: holding the demand of every scenario fixed, the controller routes only a nested random subset (10%, 25%, 50%, 75%, or 100%) of the 450 corridor vehicles, while the remainder follow a fixed free-flow shortest path to the shared destination, emulating unequipped traffic. Dijkstra and MAPPO route the same subset, and fleet metrics are computed over all 450 corridor vehicles, so results are comparable across penetration levels; the 100% level is exactly the deployment configuration above.

Reproducibility. Code, checkpoints, logs, and scripts regenerating every number and figure accompany this submission anonymized, and will be released publicly at camera-ready.

5 Results and Discussion

5.1 Training Dynamics

Figure 2 summarizes the 150-episode run. Episode outcomes are dominated by the demand realization, not by smooth policy improvement: fleet travel time swings from 254.3 s (episode 59) to 1174.8 s (episode 63) and correlates strongly with teleports ($r=0.81$): the seeds that draw heavy demand gridlock regardless of routing (panels a,b). Completion stays high (mean 0.996). Underneath the noise the policy sharpens: entropy anneals from 0.98 to 0.62 (panel c) and the mean margin between candidate scores grows from near zero to 0.89. The policy detours throughout training (per-episode non-shortest rate 30–60%), so the routing signal is present in the weights the whole time. However, the PPO update magnitude stays small (the approximate KL divergence has a median of only 0.0019 against a 0.015 target and never trips early stopping, panel d), so the per-update signal is weak: with many

Table 2: Deployment over 20 Held-Out Scenarios (Seeds 4010–4029)

Metric	Dijkstra	MAPPO (guard on)	MAPPO (guard off)
Completion rate	1.000	1.000	1.000
Avg. travel time (s)	596.6	534.3	516.9
vs. Dijkstra	–	–10.4%	–13.4%
p50 travel time (s)	576.7	532.8	504.9
p90 travel time (s)	1097.6	972.2	939.1
p95/p50 ratio	2.17	2.07	2.12
Deadline misses	225.4	208.8	206.6
Non-shortest rate	–	0.38	0.40
Scenarios won	–	15/20	15/20
Wilcoxon p vs. Dij.	–	0.019	0.008

near-equal parallel routes on a grid, the advantage of any single choice is small and easily buried by traffic variance. Value-target normalization keeps the critic well-conditioned (median value loss 0.51). Because later episodes are not reliably faster and the fleet detour *rate* drifts slowly upward (Section 5.6), we select the checkpoint by held-out greedy evaluation rather than final weights: this picked episode 74, which on the 30-seed selection block (6000–6029) completes every vehicle and runs 27.0% below Dijkstra (440.2 s vs. 603.5 s, 22/30 wins, p90 –24.2%) at a 38.6% non-shortest rate—the recalibrated guard now executes rather than reverts the learned detours. That block draws several catastrophic-congestion seeds the policy roughly halves (e.g. seed 6018 1542 → 685 s), so its gains run larger than on the disjoint deployment block next.

5.2 Deployment: MAPPO vs. Dijkstra

We deploy the selected checkpoint on a *disjoint* block of 20 held-out scenarios (seeds 4010–4029, never used for training or checkpoint selection) against Dijkstra on identical demand (Table 2, Fig. 3). Both complete 100% of vehicles. Greedy MAPPO (with the detour guard active, our default deployment) reduces mean fleet travel time from 596.6 s to 534.3 s (–10.4%, median per-scenario –33.0 s), winning 15 of 20 scenarios (paired Wilcoxon $p=0.019$), and improves p90 travel time (972 s vs 1098 s, –11.4%), the p95/p50 spread (2.07 vs 2.17), and deadline misses (209 vs 225). Here the policy executes a non-shortest route 37.6% of the time, combining live replanning with genuine load-spreading detours. The improvement grows with congestion ($r= -0.57$ between per-scenario gain and Dijkstra travel time, Fig. 3a): the largest wins are the most congested scenarios (seed 4028 1109 → 702 s, seed 4010 1585 → 1282 s), while the few losses are light-traffic scenarios where the shortest path is already near-optimal (seeds 4020, 4014 at +18–38 s). One scenario, seed 4012, is a genuine failure: MAPPO nearly doubles travel time (664 → 1031 s), a rare gridlock we return to in Section 5.6.

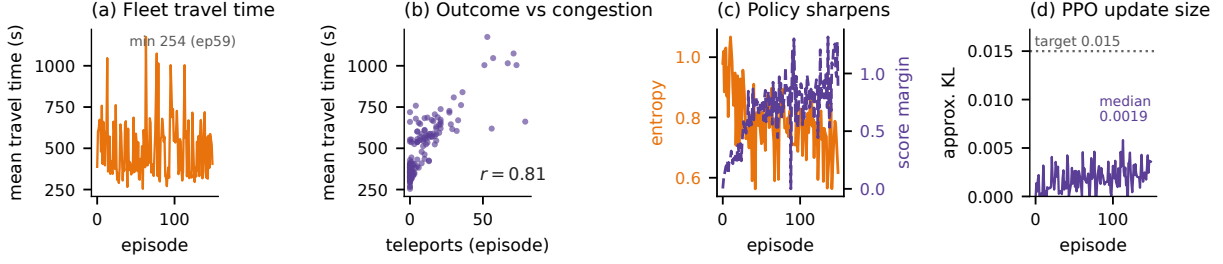


Figure 2: MAPPO training over 150 episodes on the congested corridor. (a) Fleet travel time is highly variable and (b) tracks congestion spikes (teleports): episode outcomes are demand-driven. (c) Policy entropy anneals and the candidate-score margin grows as the policy sharpens, but (d) PPO updates stay well below the target KL, indicating a weak per-update learning signal on a dense grid with many near-equal routes.

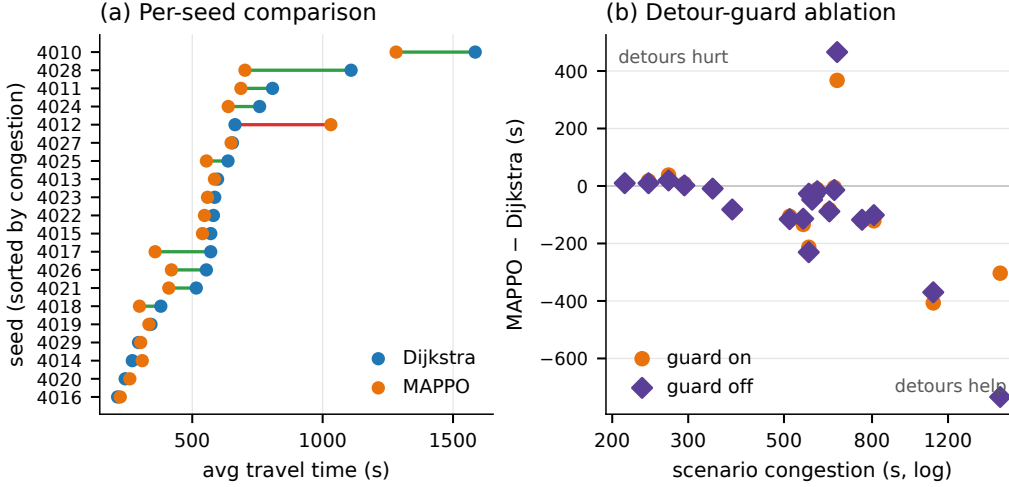


Figure 3: Deployment on 20 disjoint held-out scenarios. (a) Per-scenario fleet travel time, Dijkstra vs. greedy MAPPO (detour guard on), sorted by congestion; green connectors mark MAPPO wins, red losses, and the largest wins fall on the most congested scenarios ($r = -0.57$ between gain and congestion). (b) Detour-guard ablation: per-scenario gain over Dijkstra vs. scenario congestion (log axis) for the deployed guard-on policy (orange, 38% detours) and the guard-off policy (violet, 40% detours). Removing the guard helps most on the most congested scenarios (guard-off points fall below guard-on on the right, e.g. the rightmost pair) and is roughly neutral in light traffic; the lone point far above zero is the seed 4012 gridlock, where both configurations fail.

5.3 When Do Selfless Detours Help? A Detour-Guard Ablation

The deployed policy already executes 37.6% non-shortest routes, so the recalibrated guard is no longer an on/off switch on selflessness: it reverts only the *pile-on* subset (detours onto an alternative that is itself near capacity and buys no measurable relief). Turning it off raises the non-shortest rate only slightly, to 39.9% (Table 2), and lowers mean fleet travel time a further 3%, to 516.9 s (-13.4% , Wilcoxon $p=0.008$), with a better p90 (939.1 s); both configurations win 15/20 and complete every vehicle.

Where that extra 3% comes from is the paper’s central result: it is concentrated almost entirely at high congestion (Fig. 3b). Splitting the 20 scenarios at their median congestion, removing the guard helps the congested half by -31 s

on average but the light half by only -4 s. The clearest case is the most congested scenario, seed 4010: guard-on already cuts it $1585 \rightarrow 1282$ s, and executing the pile-on detours the guard would revert cuts it much further, to 851 s—while in light traffic the two configurations sit within noise. The guard earns its keep only on the seed 4012 gridlock, where the extra detours make things slightly worse ($1031 \rightarrow 1130$ s): a small robustness margin rather than a large peak-versus-robustness trade.

The value of selfless routing is therefore not a fixed property of the policy; it tracks the network’s price of anarchy: where contention is high the learned detours move vehicles onto genuine spare capacity and cut fleet travel time, while on a lightly loaded grid selfish shortest-path routing already approximates the system optimum and every detour is wasted distance. The recalibrated guard suppresses exactly the de-

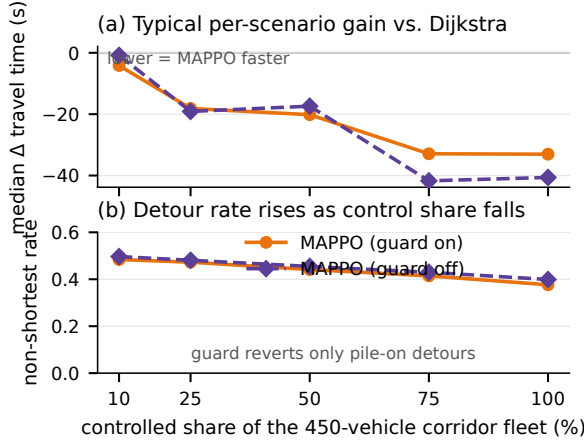


Figure 4: Fixed-demand penetration sweep on the 20 held-out scenarios. (a) Median per-scenario fleet travel-time change vs. Dijkstra: the typical gain grows monotonically with penetration. (b) With the guard removed, the policy executes non-shortest routes more often the smaller its share of the fleet.

tours that relieve nothing.

5.4 Effect of Controlled-Vehicle Penetration

Figure 4 reports the fixed-demand penetration sweep; completion stays at 100% at every level. The typical benefit of the deployed policy grows monotonically with the controlled share: the median per-scenario gain over Dijkstra improves from -4.1 s at 10% penetration to -18.1 s at 25%, -20.2 s at 50%, and -33 s at 75–100%, with the win rate rising from 11/20 to 15–16/20. The improvement becomes statistically reliable once the policy controls at least a quarter of the route-choosing traffic (paired Wilcoxon $p \leq 0.02$ at 25% and 75–100%): coordination needs mass before it can reliably move fleet-level outcomes. Two second-order effects qualify the trend. First, the policy *chooses* detours more often the less it controls (non-shortest rate 38% at full penetration rising to 48% at 10%, Fig. 4b), consistent with its observations containing more congestion that no teammate will clear. Second, mean (as opposed to median) gains are erratic at intermediate penetration, where mixed live/fixed traffic—never seen during training—produces gridlock outliers (the seed 4012 mode of Section 5.6); robust partial adoption therefore calls for training under mixed autonomy, not just deploying into it.

5.5 Greedy vs. Stochastic Deployment

Because the recalibrated guard now lets most detours through, greedy (argmax) and stochastic (sampled) deployment no longer coincide. On the 30 selection seeds (three samples each, guard active), sampling spreads slightly more load (42% vs. 39% non-shortest) but is net slower—477.5 s vs. the greedy 440.2 s, a paired advantage for greedy on nearly every seed (sign test $p=0.001$)—both completing every vehicle. The argmax already captures the useful detours

while extra sampling mostly adds pile-on choices, so we deploy greedily; route diversity comes from the policy and the reservation field, not sampling noise.

5.6 Limitations

Four limitations qualify these results. First, the aggregate gains are driven by a minority of severe-congestion scenarios; this is arguably the point of selfless routing, but it inflates means and is why the more-congested selection block (-27%) improves far more than the disjoint deployment block (-10.4%). Second, one scenario (seed 4012) is a true failure in which the greedy policy gridlocks and even the guard’s pile-on veto does not prevent it: the coordination guardrails are necessary but not sufficient. Third, because the reward settles each agent’s detour credit on decision-time estimates, the fleet detour *rate* drifts slowly upward (bounded near 44%, not divergent), which we currently contain by best-checkpoint early stopping rather than by a stable attractor—a detour-rate penalty or counterfactual critic could supply one. Fourth, the weak per-update learning signal (Section 5.1) means the policy is likely under-trained, and the moderate price of anarchy of a dense grid still caps how much any routing policy can gain, motivating higher-price-of-anarchy networks. The honest contribution is the *conditions* under which selfless routing helps, not a single benchmark number.

6 Conclusion and Future Work

We studied selfless traffic routing in SUMO with a cooperative MAPPO policy: a centralized critic, a shared per-candidate route actor, and a difference-reward team objective in seconds of travel time, with a congestion-gated incentive that pays for a detour only when it relieves a real jam. On a congested New York City corridor, deployed greedily on 20 disjoint held-out scenarios, it completes every vehicle and cuts mean fleet travel time by 10.4% (15/20 wins, $p=0.02$), with a lower tail and fewer deadline misses—executing genuine load-spreading detours nearly 40% of the time, not by replanning alone.

The more interesting finding is conditional. Ablating the guard so that even the pile-on detours execute helps a further few percent, but almost all of it falls on the most congested scenarios and vanishes in light traffic. Selfless routing pays in proportion to the network’s price of anarchy—so the gain is largest where congestion is worst (up to -27% on the most congested block), and a penetration sweep shows it growing monotonically with the controlled share, reliable once the policy routes at least a quarter of the fleet.

Several directions follow. Most directly motivated by our data is evaluation on high-price-of-anarchy networks (Pigou- or Braess-style bottlenecks), where selfless routing has a large, persistent margin rather than the intermittent one a dense grid offers. The weak learning signal and the detour-rate drift invite stronger credit assignment (a counterfactual critic (Foerster et al. 2018) or marginal-cost pricing), and partial adoption calls for training under mixed-autonomy traffic. Finally, a graph-neural-network encoding would let a policy transfer across networks, toward a deployable controller.

References

- Braess, D.; Nagurney, A.; and Wakolbinger, T. 2005. On a Paradox of Traffic Planning. *Transportation Science*, 39(4): 446–450.
- Dai, G.; Paluri, P. K.; Carmichael, T.; Cheng, A. M. K.; and Miikkulainen, R. 2019. Work-in-Progress: Leveraging the Selfless Driving Model to Reduce Vehicular Network Congestion. In *2019 IEEE Real-Time Systems Symposium (RTSS)*, 548–551.
- Dijkstra, E. W. 1959. A Note on Two Problems in Connexion with Graphs. *Numerische Mathematik*, 1(1): 269–271.
- Foerster, J.; Farquhar, G.; Afouras, T.; Nardelli, N.; and Whiteson, S. 2018. Counterfactual Multi-Agent Policy Gradients. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 32.
- IEEE. 2022. IEEE Standard for Information Technology—Amendment 5: Enhancements for Next Generation V2X. IEEE Standard 802.11bd-2022. [Online]. Available: <https://standards.ieee.org/ieee/802.11bd/7451/>.
- Iliopoulos, C.; Iossifides, A.; Foh, C. H.; and Chatzimisios, P. 2025. IEEE 802.11bd for Next-Generation V2X Communications: From Protocol to Services. *IEEE Communications Standards Magazine*, 9(2): 88–98.
- Kingma, D. P.; and Ba, J. 2015. Adam: A Method for Stochastic Optimization. In *International Conference on Learning Representations (ICLR)*.
- Krajzewicz, D.; Erdmann, J.; Behrisch, M.; and Bieker, L. 2012. Recent Development and Applications of SUMO – Simulation of Urban MObility. *International Journal on Advances in Systems and Measurements*, 5(3–4): 128–138.
- Lopez, P. A.; Behrisch, M.; Bieker-Walz, L.; Erdmann, J.; Flötteröd, Y.-P.; Hilbrich, R.; Lücken, L.; Rummel, J.; Wagner, P.; and Wießner, E. 2018. Microscopic Traffic Simulation using SUMO. In *2018 21st International Conference on Intelligent Transportation Systems (ITSC)*, 2575–2582.
- Lowe, R.; Wu, Y.; Tamar, A.; Harb, J.; Abbeel, P.; and Mordatch, I. 2017. Multi-Agent Actor-Critic for Mixed Cooperative-Competitive Environments. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 30.
- Mnih, V.; Kavukcuoglu, K.; Silver, D.; Rusu, A. A.; Veness, J.; Bellemare, M. G.; Graves, A.; Riedmiller, M.; Fidjeland, A. K.; Ostrovski, G.; Petersen, S.; Beattie, C.; Sadik, A.; Antonoglou, I.; King, H.; Kumaran, D.; Wierstra, D.; Legg, S.; and Hassabis, D. 2015. Human-level Control Through Deep Reinforcement Learning. *Nature*, 518(7540): 529–533.
- OpenStreetMap contributors. 2026. OpenStreetMap. <https://www.openstreetmap.org>. Map data licensed under the Open Database License (ODbL).
- Roughgarden, T.; and Tardos, É. 2002. How Bad Is Selfish Routing? *Journal of the ACM*, 49(2): 236–259.
- Schulman, J.; Moritz, P.; Levine, S.; Jordan, M.; and Abbeel, P. 2016. High-Dimensional Continuous Control Using Generalized Advantage Estimation. In *International Conference on Learning Representations (ICLR)*.
- Schulman, J.; Wolski, F.; Dhariwal, P.; Radford, A.; and Klimov, O. 2017. Proximal Policy Optimization Algorithms. *arXiv preprint arXiv:1707.06347*.
- Wardrop, J. G. 1952. Some Theoretical Aspects of Road Traffic Research. *Proceedings of the Institution of Civil Engineers*, 1(3): 325–362.
- Wolpert, D. H.; and Tumer, K. 2001. Optimal Payoff Functions for Members of Collectives. *Advances in Complex Systems*, 4(2–3): 265–279.
- Yu, C.; Velu, A.; Vinitzky, E.; Gao, J.; Wang, Y.; Bayen, A.; and Wu, Y. 2022. The Surprising Effectiveness of PPO in Cooperative Multi-Agent Games. *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, 35: 24611–24624.
- Zhang, J.; Pourazarm, S.; Cassandras, C. G.; and Paschalidis, I. C. 2018. The Price of Anarchy in Transportation Networks: Data-Driven Evaluation and Reduction Strategies. *Proceedings of the IEEE*, 106(4): 538–553.